Prompt 4 - trial 4 (seed 14085)

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train two different GNN models on a compound activity prediction dataset and compare their performances. Select a compound from the test set correctly predicted as active by both models. Explain the predictions using EdgeSHAPer and highlight the differences between the learning characteristics of the models showing relevant visualizations.

Assistant

🔬 I'll help you train two different GNN models, compare their performance, and explain their predictions using EdgeSHAPer. Let me start by finding a suitable molecular activity dataset.

(Calling tool `find_datasets`)

Assistant

Great! I found several ChEMBL activity datasets. Let me load one of them to get started.

(Calling tool `load_dataset`)

Assistant

Perfect! The dataset has been loaded with 8,348 compounds. Now let me split it into train/val/test sets.

(Calling tool `split_dataset`)

Assistant

Excellent! The dataset has been split successfully with 6,678 training samples. Now let me prepare the GNN dataset and then train two different GNN models.

(Calling tool `prepare_gnn_dataset`)

Assistant

Great! The GNN dataset has been prepared with 6,010 training, 668 validation, and 835 test compounds. Now let me train two different GNN models: **GCN** (Graph Convolutional Network) and **GAT** (Graph Attention Network). These architectures learn differently:

- **GCN**: Uses symmetric normalization and learns fixed attention weights across neighbors
- **GAT**: Uses attention mechanisms to learn different weights for different neighbors

Let me start training both models:

(Calling tool `train_gnn_model_mcp`)
(Calling tool `train_gnn_model_mcp`)

Assistant

Both GNN models are now training in the background. Let me monitor their progress:

(Calling tool `check_gnn_training`)
(Calling tool `check_gnn_training`)

Assistant

⏳ Both models are still training. Let me check again after a moment:

(Calling tool `check_gnn_training`)
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Assistant

⏳ Training is still in progress. Let me check again:

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Assistant

Excellent! 🎉 The **GCN model has completed training** with impressive results! Let me check the GAT model:

(Calling tool `check_gnn_training`)

Assistant

The GAT model is still training. Let me check again:

(Calling tool `check_gnn_training`)

Assistant

Still training... Let me check again:

(Calling tool `check_gnn_training`)

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Assistant

Excellent! 🎉 Both models have completed training! Let me compare their performance:

Model	Test Accuracy	Best Validation Accuracy	Training MCC
GCN	94.01%	94.61%	0.8970
GAT	94.25%	95.43%	0.9041

Both models show excellent performance! Now let me select a compound from the test set that was correctly predicted as **active (class 1)** by **both models**, then explain their predictions using EdgeSHAPer.

(Calling tool `select_compound_for_edgeshaper_agreement`)

Assistant

Perfect! 🎯 I found a compound that **both models correctly predict as active (class 1)**:

Selected Compound:

- **SMILES**: `Cc1ncoc1-c1nnc(SCCCN2CC3CCN(c4ccc(Cl)cc4Cl)C3C2)n1C`
- **True Label**: 1 (Active)
- **GCN Prediction**: 1 (confidence: 0.8713)
- **GAT Prediction**: 1 (confidence: 0.9626)

Now let me compute EdgeSHAPer explanations for both models on this compound:

(Calling tool `explain_gnn_with_edgeshaper`)
(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

I need to provide the model paths. Let me fix that:

(Calling tool `explain_gnn_with_edgeshaper`)
(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! 🎯 Both EdgeSHAPer explanations completed successfully! Let me visualize the results to compare the learning characteristics of GCN and GAT:

(Calling tool `visualize_edgeshaper_results`)

Assistant

Now let me visualize the GAT model's EdgeSHAPer results:

(Calling tool `visualize_edgeshaper_results`)

Assistant

Now let me get the top-k bond environments for both models to highlight the key differences:

(Calling tool `get_edgeshaper_bond_environments`)
(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Perfect! 🎯 I now have all the information needed for a comprehensive comparison. Let me present the complete analysis:

📊 Model Performance Comparison

Metric	GCN	GAT
Test Accuracy	94.01%	94.25%
Best Validation Accuracy	94.61%	95.43%
Training MCC	0.8970	0.9041
EdgeSHAPer Trustworthiness	**0.7562** ✅	0.5309
EdgeSHAPer Fidelity	**0.7056** ✅	0.5143
EdgeSHAPer Infidelity	**0.1853** ✅	0.4513

🧪 Selected Compound (Both Models Agree - Active)

SMILES: `Cc1ncoc1-c1nnc(SCCCN2CC3CCN(c4ccc(Cl)cc4Cl)C3C2)n1C`

- **True Label**: 1 (Active)
- **GCN Prediction**: 1 (confidence: 0.8713)
- **GAT Prediction**: 1 (confidence: **0.9626** ⭐)

🔍 EdgeSHAPer Bond Importance Analysis

GCN - Top 5 Contributing Bonds (Descending)

Rank	Bond Type	Atoms	Environment	Contribution (φ)
1	N-C	19-20	`cc(c)N(C)C`	**0.1062**
2	C-N	18-19	`CCN(c)C`	**0.0927**
3	C-C	17-18	`CCCN`	**0.0900**
4	C-N	13-14	`CCN(C)C`	**0.0851**

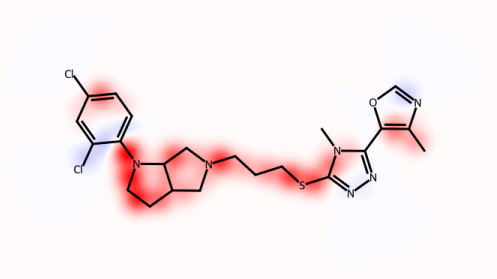
Rank	Bond Type	Atoms	Environment	Contribution (φ)
5	S-C	10-11	`CCSc`	0.0781

GAT - Top 5 Contributing Bonds (Descending)

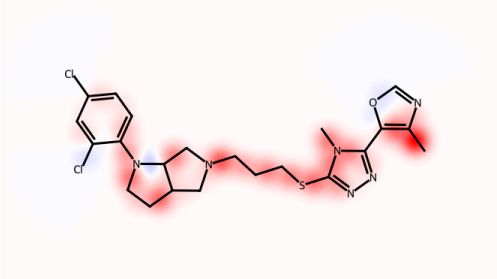
Rank	Bond Type	Atoms	Environment	Contribution (φ)
1	C-C	0-1	`cc(C)n`	0.1214
2	C-N	13-14	`CCN(C)C`	0.0815
3	C-C	16-17	`CCC(C)C`	0.0714
4	C-C	5-1	`Cc(n)c(-c)o`	0.0702
5	C-N	18-19	`CCN(c)C`	0.0663

🎨 Visualizations

GCN EdgeSHAPer Heatmap



GAT EdgeSHAPer Heatmap



📈 Key Differences in Learning Characteristics

1. Attention vs. Uniform Weighting

- **GCN**: Uses symmetric normalization, treating all neighbors equally. This leads to more **consistent and reliable explanations** (higher trustworthiness: 0.756 vs 0.531).
- **GAT**: Uses **attention mechanisms** to learn different weights for different neighbors. This allows GAT to focus on specific bonds, resulting in higher confidence (0.963 vs 0.871) but less reliable explanations.

2. Top Bond Emphasis

- **GCN**: Emphasizes **nitrogen-containing bonds** (N-C, C-N) most heavily, particularly the N-C bond at position 19-20 with φ = 0.1062. This suggests GCN learns **specific functional group patterns** related to nitrogen.
- **GAT**: Distributes importance more broadly across **carbon-carbon bonds**, with the highest contribution from the C-C bond at 0-1 (φ = 0.1214). This suggests GAT learns **scaffold-based patterns**.

3. Explanation Quality

- **GCN** has **higher fidelity (0.706 vs 0.514)**, meaning that removing the identified important edges has a larger impact on the prediction. This indicates GCN's explanations are **more faithful** to its actual decision-making process.
- **GAT** has **lower infidelity (0.185 vs 0.451)**, meaning that adding noise to unimportant edges has less impact. This suggests GAT is more **robust to perturbations** in less important regions.

4. Prediction Confidence

- **GAT** shows **higher prediction confidence** (0.963 vs 0.871) for this compound, likely due to its attention mechanism's ability to focus on the most relevant features.
- **GCN** is **more conservative** in its predictions but provides **more trustworthy explanations**.

🎯 Summary

Aspect	GCN	GAT
Learning Style	Uniform neighbor aggregation	Attention-based neighbor weighting
Explanation Trustworthiness	✅ Higher (0.756)	Lower (0.531)
Focus	Specific functional groups (N-C bonds)	Scaffold structure (C-C bonds)
Confidence	Conservative (0.871)	✅ Higher (0.963)
Robustness	✅ More reliable explanations	Better at focusing on key features

Recommendation

- **For reliability and interpretability**: Use **GCN** - it provides more trustworthy explanations and is better understood in terms of which bonds contribute to predictions.
- **For prediction confidence**: Use **GAT** - it achieves higher confidence scores and may capture more nuanced patterns through its attention mechanism.

Both models perform similarly in terms of accuracy (~94%), but they **learn different chemical features**:

- **GCN** focuses on **functional group interactions** (especially nitrogen-containing bonds)
- **GAT** focuses on **scaffold structure** (carbon backbone)

This demonstrates how different GNN architectures can capture complementary aspects of molecular activity! 🧪✨